

Introduction to Data Science (CS2205)

Books:

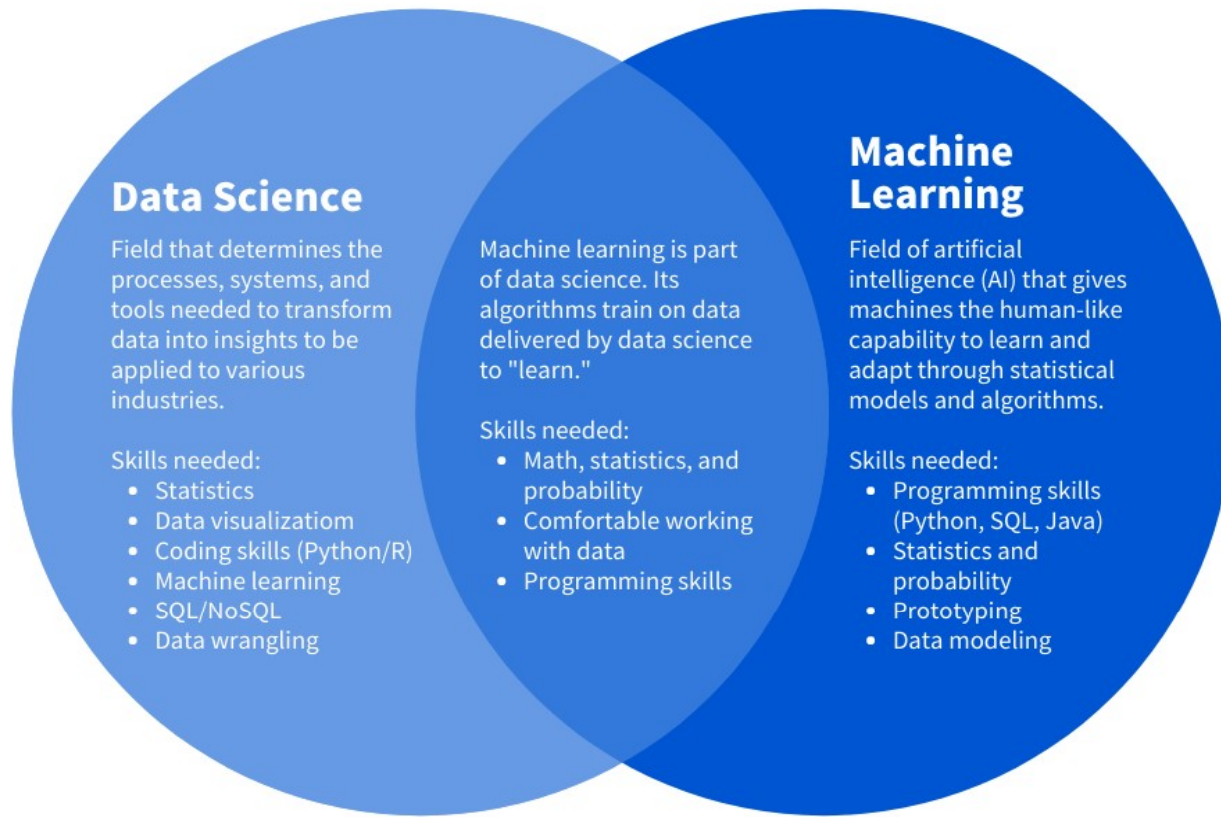
1. Foundations of Data Science by Avrim Blum, John Hopcroft and Ravindran Kannan.
2. Data Mining: Concepts and Techniques by Jiawei Han, Micheline Kamber and Jian Pei.
3. Introduction to Data Mining by Pang-Ning Tan, Michael Steinbach and Vipin Kumar
4. Machine Learning: A Probabilistic Perspective by Kevin P. Murphy
5. Mining of Massive Datasets by Jure Leskovek, Anand Rajaraman and Jeffrey Ullman
6. Data Science for Business: What You Need to Know about Data Mining and Data-analytic Thinking by Foster Provost and Tom Fawcett.

What is Data Science?

- Data Science combines several aspects of Data such as Technology, Algorithm development, and data interference to study the data, analyze it, and find innovative solutions to difficult problems.
- Data Science is all about Analyzing data and driving for business growth by finding creative ways.
- In other words, we had a lot of data with us, but we were not able to find out any insights from it. The need to understand and analyze data to make better decisions is what gave birth to Data Science.

Is Machine Learning and Data Science same?

- While data science and machine learning are related, they are very different fields.
- The two relate to each other in a similar way that squares are rectangles, but rectangles are not squares. Data science is the all-encompassing rectangle, while machine learning is a square that is its own entity.
- In a nutshell, data science brings structure to big data while machine learning focuses on learning from the data itself.
- Data science focuses on managing, processing, and interpreting big data to effectively inform decision-making. Machine learning leverages algorithms to analyze data, learn from it, and forecast trends.



- In recent years, machine learning and artificial intelligence (AI) have dominated parts of data science, playing a critical role in data analytics and business intelligence.
- Data Science and Machine Learning are often used by data scientists in their work and are rapidly being adopted by nearly every industry.

Applications of Data Science

- **Healthcare:** The application of data science across various sectors is growing tremendously. Healthcare is one of the major sectors that is being transformed gradually by data science.
- **Internet Search:** Many search engines like Yahoo, Google, and Bing use data science algorithms to deliver the best output for our search query within the second.

Applications of Data Science

- **Fraud Detection:** The data is drawn randomly from the various sectors and platforms like surveys on phones, emails, social media platforms, etc. Data science gives creative, scientific, and investigative thinking to big data.
- **Image Recognition:** In this digital era, data science tools have started recognizing the human face with all the pictures available in its database.

Data Science Life Cycle

- The data science life cycle is a complex and iterative process that involves six phases:
 - i. Problem identification
 - ii. Data collection
 - iii. Data preparation
 - iv. Data modeling and analysis
 - v. Model evaluation, and
 - vi. Deployment.
- This comprehensive approach requires the collaborative efforts of various professionals who possess different skills and experiences.

Persons involved in Data Science Life Cycle

- The success of a data-driven project depends on the synergy between various professionals participating in the data science life cycle.
- Each role is equally important and influences the final outcome of any project.
- To successfully implement data-driven solutions, it's crucial to build teams that share different roles and responsibilities.

Domain Expert

- A domain expert is responsible for understanding the industry-specific challenges and opportunities that can be addressed using data-driven solutions.
- They possess deep knowledge of their field and collaborate with other team members to identify business-related issues that can be solved by analyzing specific datasets.
- Domain experts play a critical role in defining the goals of a project and validate the rationale of developed models.

Data Scientist

- Data scientists are skilled professionals who perform advanced analytics on structured and unstructured datasets to drive actionable insights.
- They use various techniques such as machine learning algorithms, statistical analysis, and predictive modeling to achieve desired outcomes.
- With strong coding skills and programming languages proficiency (such as Python or R), data scientists transform raw data into useful insights that help businesses to make decisions.
- They also visualize their findings using compelling visualizations to facilitate better understanding among stakeholders. There are many visualization tools, like Tableau, Microsoft Power BI, Datawrapper, Infogram, Qlik Sense, etc.

Data Engineer

- Data engineers are responsible for developing, maintaining, and optimizing the pipelines that streamline the flow of data between various sources.
- With their strong software engineering background, they create robust infrastructures that can handle massive amounts of raw information.
- This enables other team members to analyze these datasets efficiently.
- They also play a vital role in the implementation of various storage methods like databases or cloud-based services to keep the company's repositories up-to-date and accessible.

Data Analyst

- Data analysts focus on interpreting collected information to address specific business-related questions.
- Their main responsibility lies in processing raw datasets and generate insights that can help in decision making.
- They employ descriptive analytics techniques to paint a clearer picture of the gathered information and often use visualization tools like Tableau or Power BI to communicate their findings.
- Data analysts work closely with domain experts and data scientists to understand the context behind the data and ensure that the extracted insights are relevant and meaningful.

Artificial Intelligence Engineer

- Artificial intelligence engineers possess expertise in designing, developing, and deploying AI models for various applications.
- With a strong background in machine learning algorithms, computer vision, natural language processing, and robotics, they build intelligent systems that can automate complex tasks or improve existing solutions.
- They are responsible for identifying opportunities where AI-based technologies can be implemented.

6 Steps of Data Science Lifecycle

- The data science lifecycle is a structured framework that guides data scientists through the process of solving real-world problems using data.
- It consists of interconnected stages, ensuring that the entire journey, from problem identification to the deployment of solutions, is streamlined and efficient.
- By following this lifecycle, data scientists can extract valuable insights and drive positive outcomes for businesses.

1 – Problem Identification and Business Understanding

- At the beginning of every data science project, it is of utmost importance to establish a clear problem statement and define the objectives of the business.
- This step involves close collaboration with stakeholders to gain a thorough understanding of their specific requirements and expectations.
- By engaging in this collaborative process, data scientists can ensure that their efforts are fully aligned with the desired outcomes and deliver solutions that address the core challenges.

1 – Problem Identification and Business Understanding

- Defining a well-articulated problem statement and understanding of the business lays the foundation for the entire data science project.
- It serves as a guiding compass which can direct focus of the analysis and provide a framework for the subsequent stages of the data science lifecycle.

2 – Data Collection and Exploration

- Once problem is clearly defined, data collection becomes a critical aspect of the data science life cycle.
- This stage entails gathering raw data from various sources like databases, spreadsheets, web scraping or APIs.
- Make sure you include possible external influences as well, such as seasonal trends and economic indicators.
- Having enough good-quality data is important for building accurate models later in the project.

2 – Data Collection and Exploration

- Moreover, while collecting data, it's crucial to maintain its originality and keep track of its provenance for transparency and reproducibility purposes.
- Data acquisition can take various forms according to nature of the problem and the specific requirements of the project.
- It may involve accessing publicly available datasets, such as government databases, open data repositories, or industry-specific data sources.
- These datasets can provide a wealth of information and serve as a foundation for analysis.

2 – Data Collection and Exploration

- Once the data has been acquired, it is imperative to explore and examine it thoroughly.
- This exploration phase involves probing the data's structure, characteristics, and potential limitations.
- Data scientists need to understand the variables, their distributions, and the relationships between them.
- Exploratory data analysis techniques, such as statistical summaries, data visualizations, and descriptive statistics, are employed to gain a deeper understanding of the data.
- Furthermore, during the exploration phase, data scientists assess the quality and integrity of the acquired data. They identify missing values, outliers, inconsistencies, or any other data anomalies that may impact the analysis.

3 – Data Preparation and Cleaning

- Data, in its raw form, is frequently riddled with inconsistencies, missing values, and other irregularities that can hinder effective analysis.
- Therefore, in the data science lifecycle, the data preparation phase plays a critical role in transforming raw data into a clean and usable format.
- This crucial step ensures that the data is reliable, accurate, and ready for analysis, setting the stage for meaningful insights to be extracted.
- During the data preparation phase, data scientists employ a range of techniques to address the various challenges posed by the raw data.

3 – Data Preparation and Cleaning

- One common task involves handling missing values, which are data points that are absent or incomplete.
- Missing values can significantly impact the accuracy of analyses, as they introduce uncertainty and potentially bias the results.
- Data scientists use strategies such as imputation, where missing values are estimated or replaced using statistical methods, to ensure that the data remains robust and representative.

3 – Data Preparation and Cleaning

- Another important aspect of data preparation is the identification and treatment of outliers.
- Outliers are data points that deviate significantly from the majority of the data. These extreme values can distort statistical analyses and lead to misleading conclusions.
- Data scientists employ techniques such as statistical tests, visualization tools, or domain knowledge to identify and handle outliers appropriately, ensuring that they do not unduly influence the results.

3 – Data Preparation and Cleaning

- Normalization of variables is another task in data preparation.
- Variables in raw data often have different scales, units, or ranges, which can affect the analysis and interpretation of results.
- Normalization involves transforming variables to a common scale or range. This step facilitates accurate modeling and enhances the reliability of subsequent analyses.

3 – Data Preparation and Cleaning

- Ensuring data integrity is a key consideration in the data preparation phase.
- Data integrity refers to the accuracy, consistency, and reliability of the data.
- Data scientists perform checks and validations to verify the integrity of the data, identifying any inconsistencies, duplications, or errors.

3 – Data Preparation and Cleaning

- The quality of the data is paramount in the data science lifecycle, as it greatly influences the accuracy and reliability of subsequent analyses.
- If the data is riddled with inconsistencies, missing values, or other irregularities, it can lead to flawed conclusions and unreliable insights.
- By conducting thorough data preparation, data scientists address these issues and create a clean and robust dataset that serves as a solid foundation for accurate analysis and informed decision-making.

4 – Data Modeling and Analysis

- In this stage of the data science lifecycle, data scientists use statistical and machine learning techniques to analyze the prepared data.
- By applying these techniques, they extract meaningful information, make accurate predictions, and gain a deeper understanding of the underlying insights within the data.
- This phase is characterized by tasks such as feature selection, model training and performance evaluation.
- All of these contribute to the successful analysis and interpretation of the data.

4 – Data Modeling and Analysis

- One essential task during this stage is feature selection.
- Data scientists carefully choose the relevant features or variables from the dataset that are most informative and influential for the analysis.
- By selecting the right set of features, they can simplify the modeling process, enhance the interpretability of results, and reduce the risk of overfitting.

4 – Data Modeling and Analysis

- Once the appropriate features are identified, data scientists proceed with model training.
- This involves feeding the prepared data into various statistical or machine learning models, such as linear regression, decision trees, or neural networks.
- During the training process, the models learn from the data and establish relationships between the input features and the target variable.
- The objective is to create models that capture the inherent patterns and relationships within the data.

4 – Data Modeling and Analysis

- Hyperparameter tuning is another crucial aspect of this stage. Many models have hyperparameters, which are parameters that govern the behavior and performance of the model but are not learned from the data.
- Data scientists fine-tune these hyperparameters to optimize the model's performance and achieve the best possible results.
- Techniques such as grid search, random search, or Bayesian optimization are commonly employed to identify the optimal combination of hyperparameter values.
- For example, hyperparameters in decision trees include criteria for splitting (e.g., Gini or entropy), maximum depth, minimum samples per leaf, and the number of features considered for splitting.

5 – Model Evaluation and Interpretation of Results

- Once the data models have undergone training and predictions have been generated, the subsequent step in the data science lifecycle is to evaluate the results.
- Data scientists meticulously assess the performance of their models and validate the accuracy of the predictions against the ground truth or known outcomes.
- This evaluation process plays a crucial role in determining the effectiveness of the models and gaining valuable insights into the analyzed data.
- During the evaluation stage, data scientists employ various techniques to analyze and interpret the results.

5 – Model Evaluation and Interpretation of Results

- Statistical analysis is a fundamental approach used to assess the performance metrics of the models.
- These metrics can include accuracy, precision, recall, F1 score, or other domain-specific measures depending on the nature of the problem.
- In addition to statistical analysis, visualizations play a significant role in evaluating the results.
- Data scientists utilize data visualization techniques to present the predictions, compare them with the actual outcomes, and identify trends, or anomalies.

5 – Model Evaluation and Interpretation of Results

- Data scientists go beyond the statistical measures and visual representations to extract actionable insights.
- By incorporating their expertise and understanding of the problem domain, data scientists can provide meaningful conclusions based on the analyzed data.
- The evaluation stage often involves iterative refinement and improvement of the models.
- If the performance metrics are not satisfactory, data scientists revisit earlier stages of the data science lifecycle to refine the data preparation, feature selection, or modeling techniques.

6 – Deployment and Communication of Findings

- In the deployment stage of the data science lifecycle, data scientists focus on translating their models and findings into real-world solutions.
- This process needs integration of models into existing systems, building interactive dashboards, or creating application programming interfaces (APIs) to facilitate easy access and utilization.
- Integrating the trained models into existing systems involves integrating analytical models into the operational infrastructure of an organization.
- By integrating the models, the organization can automate decision-making processes, optimize resource allocation, or improve operational efficiency based on the insights gained from the data analysis.

6 – Deployment and Communication of Findings

- In addition to system integration, data scientists may also develop interactive dashboards or visualization tools to present the findings in user-friendly manner.
- Dashboards provide stakeholders with a consolidated view of the insights derived from the data analysis.
- They enable users to interact with the data, explore different visualizations, and gain a deeper understanding of the underlying trends.
- Creating APIs is another common approach in the deployment stage.
- Data scientists develop APIs to provide convenient access to the models and insights.
- APIs facilitate seamless integration of the data-driven solutions into various platforms.

6 – Deployment and Communication of Findings

- Effective communication of the findings to stakeholders is a critical aspect of the deployment stage.
- Data scientists ensure that the insights derived from the data analysis are clearly and comprehensively communicated to the intended audience.
- This involves translating complex technical concepts into actionable insights that can be easily understood by stakeholders from diverse backgrounds.
- Moreover, the deployment stage may also involve monitoring the performance of the deployed models in real-world scenarios.
- Data scientists continuously evaluate the performance of the models, assess their accuracy, and identify areas for improvement.
- In this way, data scientists can ensure that the deployed solutions remain effective and relevant over time.

Thank you